



Welcome to

AASD 4011: Deep Learning II

Credit hours: 48

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George Brown College

Agenda:

1. Course Outline
2. Convolutional Neural Networks (CCN)
3. Hand on practice

Background:

AASD 4014 Deep Learning II

Convolutional NN

Image classification

Object recognition, detection

Object Segmentation, Object Classification

Object tracking

Keras, Tensorflow, Opencv

...

Major group project and a presentation due at the last week of class

Evaluation:

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	% of Final Grade:
Participation	In class participation in various activities	1, 2, 3, 4, 5	1,2,3,4,5,6,7,8,9,10,11	1-3	10
Exercises/Labs, presentation	Weekly Hands-on exercises/ Assignments	1, 2, 3, 4, 5	1,4,5,6,7,9,10,11	1-3	20
Final Exam/Project		1,2, 3, 4, 5, 6,	1,2,4,5,6,7,8,9,10,11,	4	70
TOTAL:					100%

GRADING SYSTEM the passing grade for this course is: D (50%)

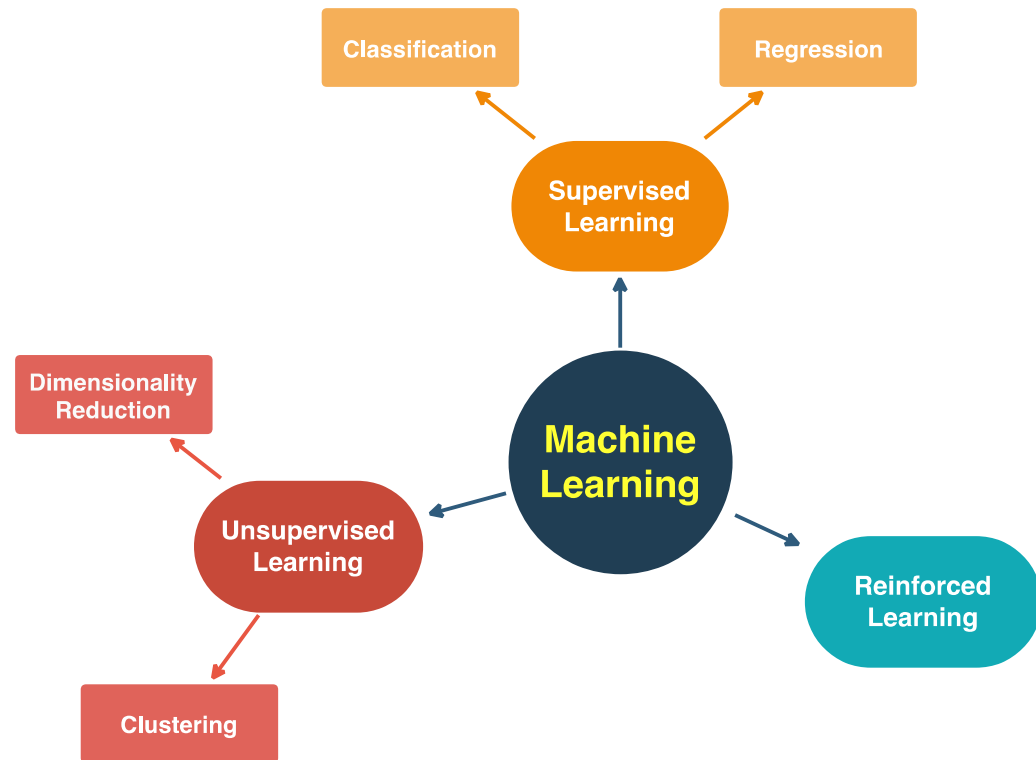
A+	90-100	4.0	B+	77-79	3.3	C+	67-69	2.3	D+	57-59	1.3	Below 50	F	0.0
A	86-89	4.0	B	73-76	3.0	C	63-66	2.0	D	50-56	1.0			
A-	80-85	3.7	B-	70-72	2.7	C-	60-62	1.7						

Machine Learning subcategories:

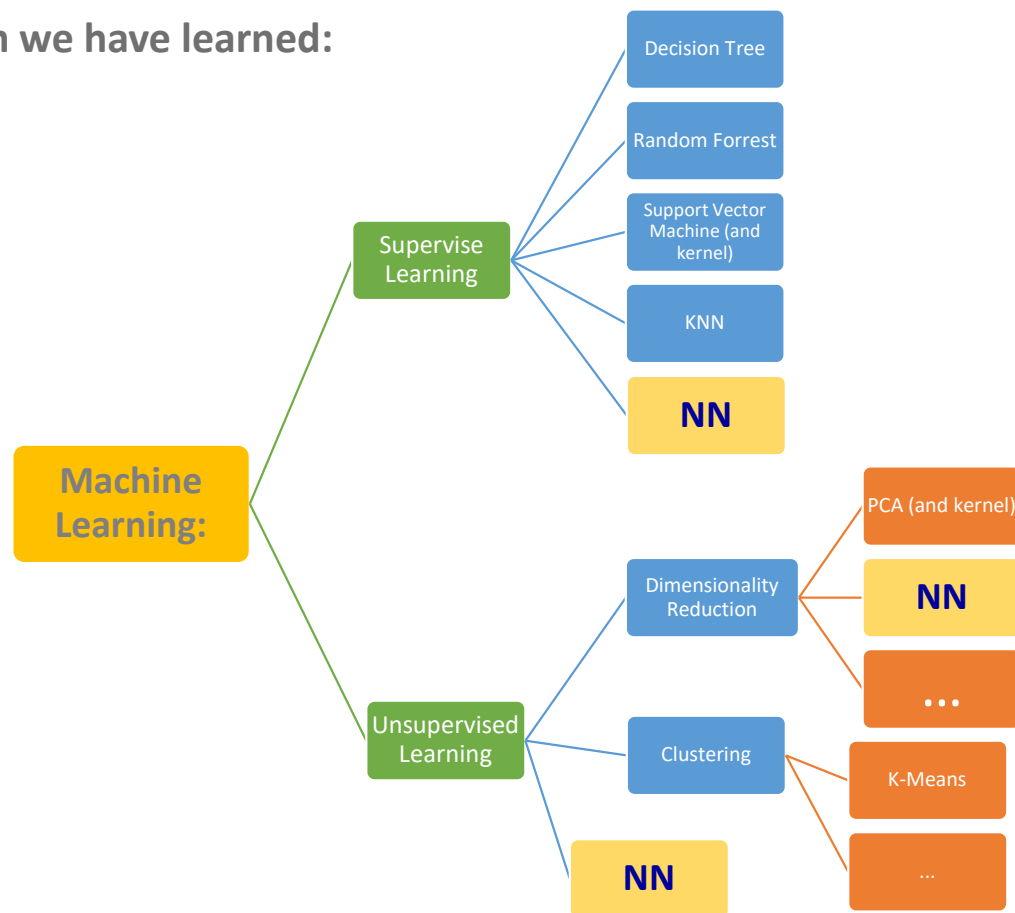
- I. Supervised Learning
- II. Un-supervised Learning
- III. Reinforced learning
- IV. Others...

Where is NN:

Almost all of them



What algorithm we have learned:



Convolutional Neural Networks

CNN definition, background:

CNN is a form of deep learning neural network that mainly deals with images.

CNN are very similar to ANN, they have layers, nodes, weights and biases.

Each node get an input, perform dot product (summation) and apply activation.

And same as ANN they have a loss function in the last layer.

Intuition behind CNN:

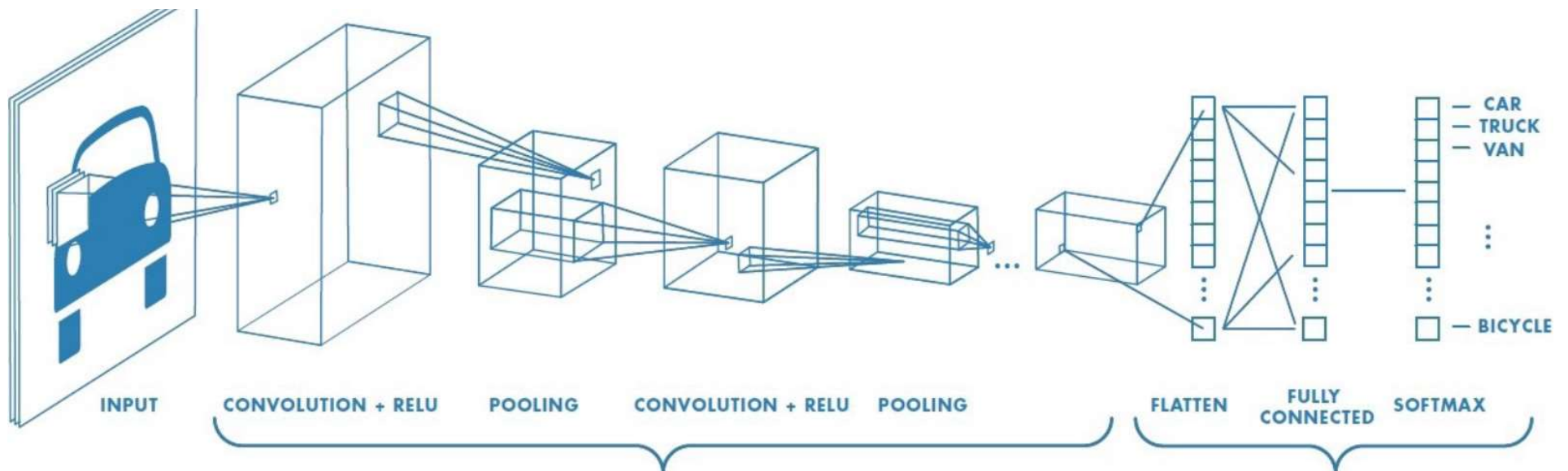
What kind of layers a CNN has:

- Input Layer
- Convolutional layer
- Pooling layer
- Flatten layer
- Fully connected layer
- Output (sigmoid/softmax)

Intuition behind CNN:

What kind of layers a CNN has:

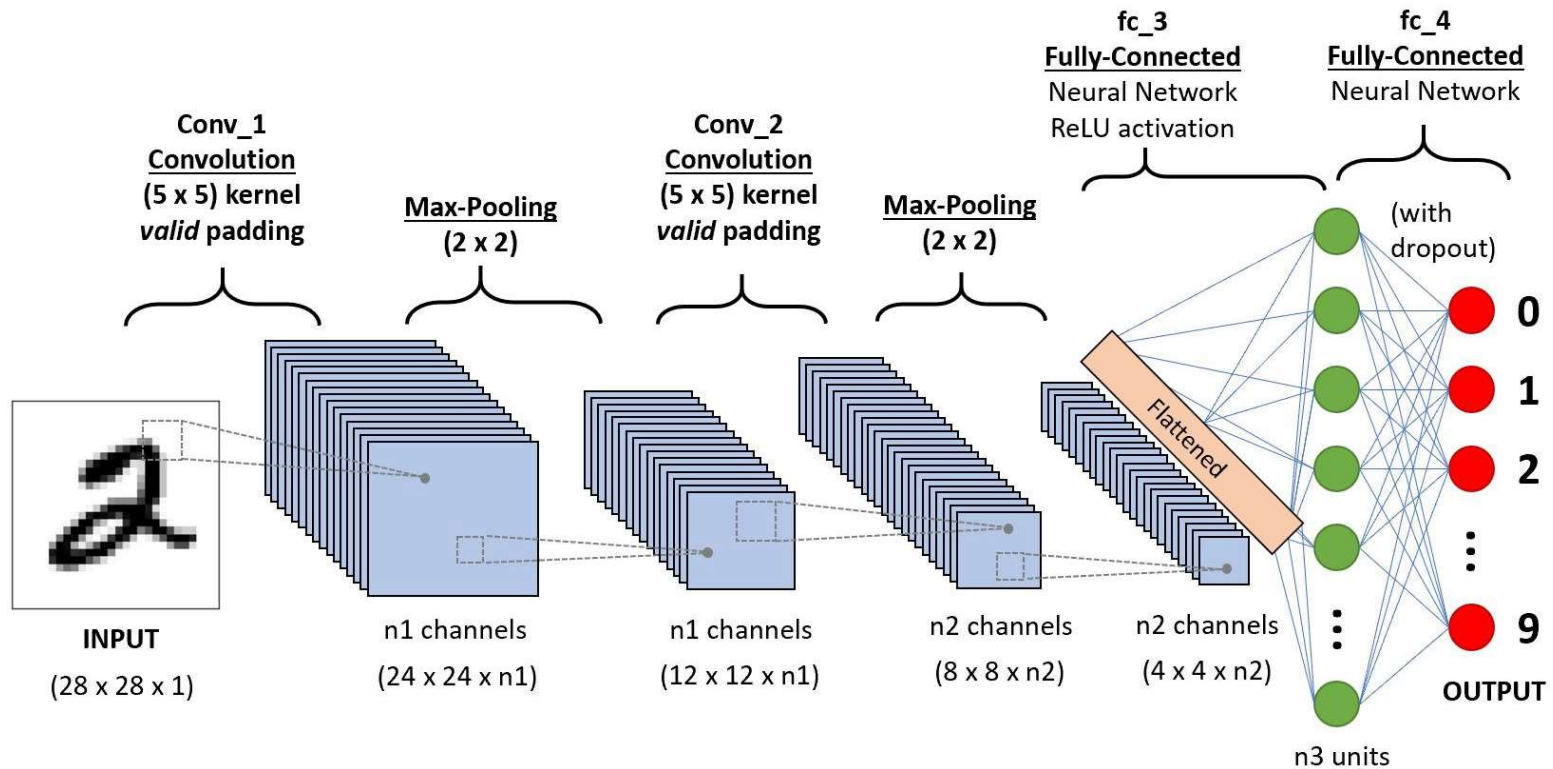
Input Layer → Convolutional layer → Pooling layer → Flatten layer → Fully connected layer → Output (sigmoid/softmax)



Intuition behind CNN:

What kind of layers a CNN has:

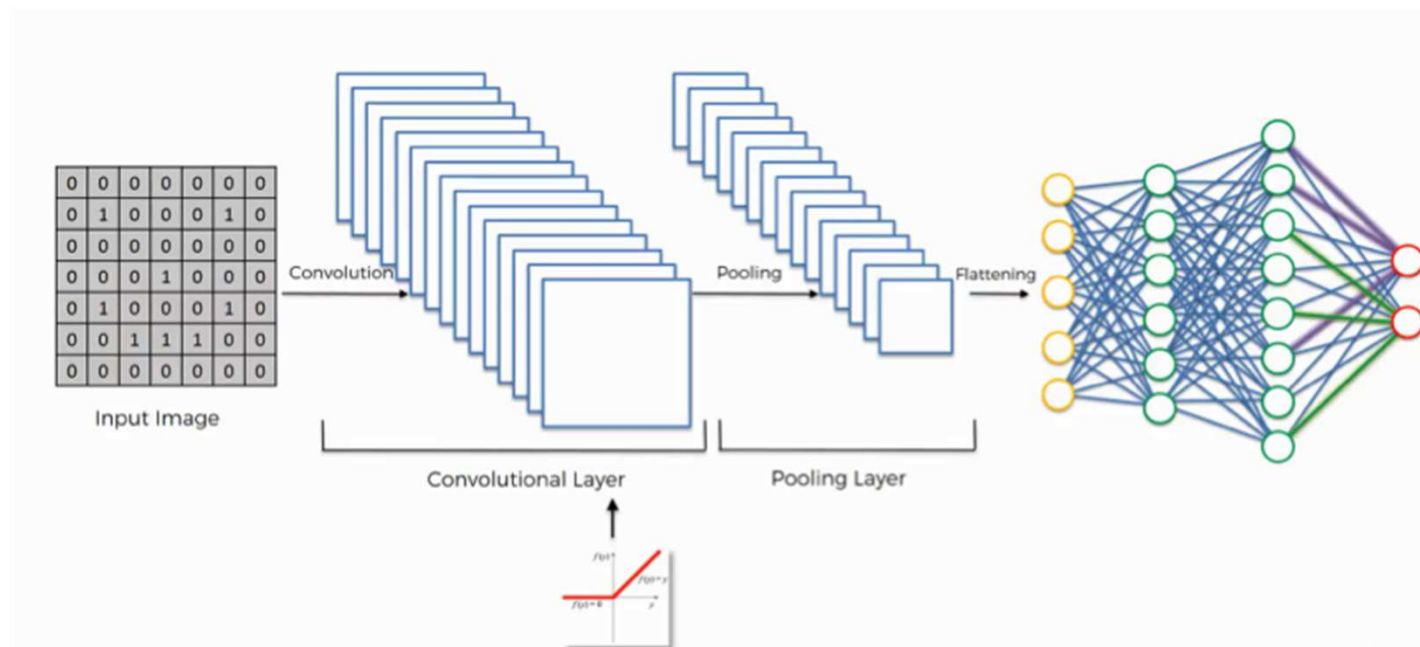
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Intuition behind CNN:

What kind of layers a CNN has:

Input Layer → Convolutional layer → Pooling layer → Flatten layer → Fully connected layer → Output (sigmoid/softmax)



Intuition behind CNN:

Different Layers:

- Input Layer

This layer has the raw pixels of the images,

Note that images all need to be the same size, if not you need to do a step before this and re-size all images to the same size.

For example for a RGB image, dimensions are [width, height, 3]

3 is the three colour channels (Red, Green, Blue)

Intuition behind CNN:

Different Layers:

- Convolution layer (Conv Layer)

This layer Convolve the input and pass it on. Think of this layer as a filter.

It has these components:

- Convolutional kernels that has a width and height. These considered as hyper-parameters.
- The number of input channels and output channels. These considered as hyper-parameters.
- The depth of the Convolution filter (the input channels)

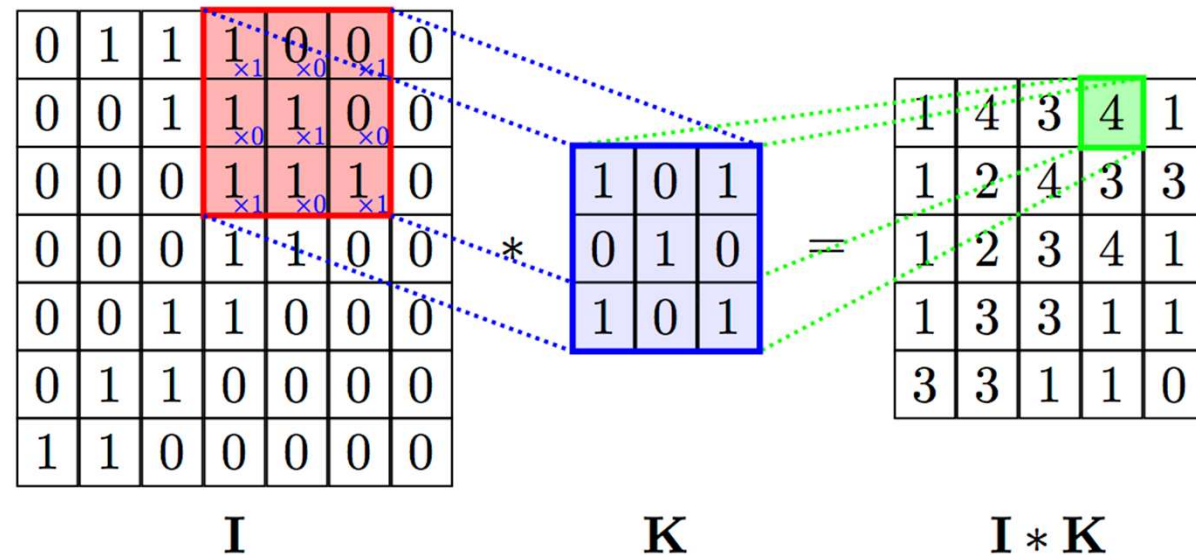
Intuition behind CNN:

Different Layers:

- Convolution layer

Convolution Kernels: The operation on two function that explain how one effect the other one

$$\begin{aligned}(f * g)(t) &\stackrel{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} f(t - \tau)g(\tau) d\tau.\end{aligned}$$



Intuition behind CNN:

Different Layers:

- Convolution layer

Convolution Kernels: another example

INPUT IMAGE					
18	54	51	239	244	188
55	121	75	78	95	88
35	24	204	113	109	221
3	154	104	235	25	130
15	253	225	159	78	233
68	85	180	214	245	0

WEIGHT		
1	0	1
0	1	0
1	0	1

429

Intuition behind CNN:

Different Layers:

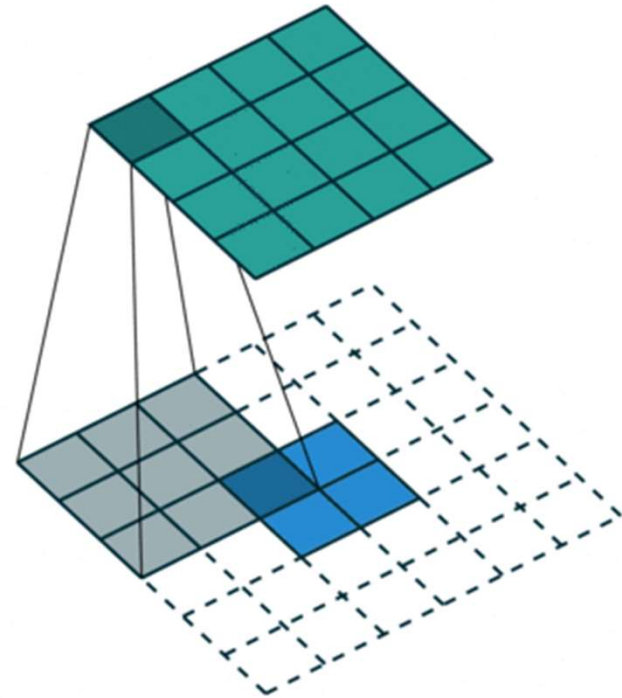
- Convolution layer

Stride: how much the kernel move after each application (how much it jumps)

Padding: addition of pixels to an image (i.e if the image size is 20×20 , after padding one pixel to right it will be 20×21)

Apply Relu: adding a Relu activation function will take care of any negative pixel value

There are several types and forms of kernels detail discussions of kernels are out of scope of the course.



Intuition behind CNN:

Different Layers:

- Convolution layer

After applying the convolution the size of the image will change (number of datapoints), lets take a look at the example we have seen before:

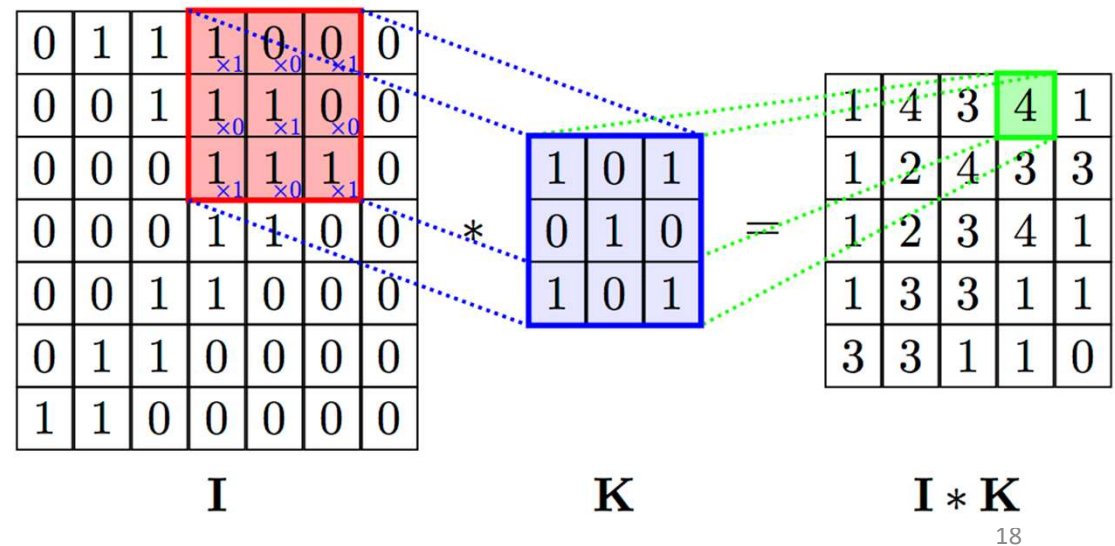
Image size: $7*7*1$

Kernel size: $3*3$

Output: $5*5*1$

Now if we apply 20 ($3*3$) kernels:

We will get $20 * (5*5)$



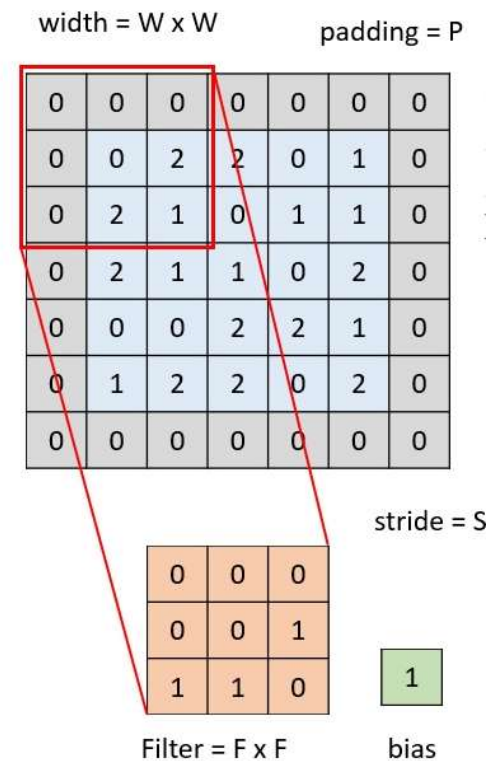
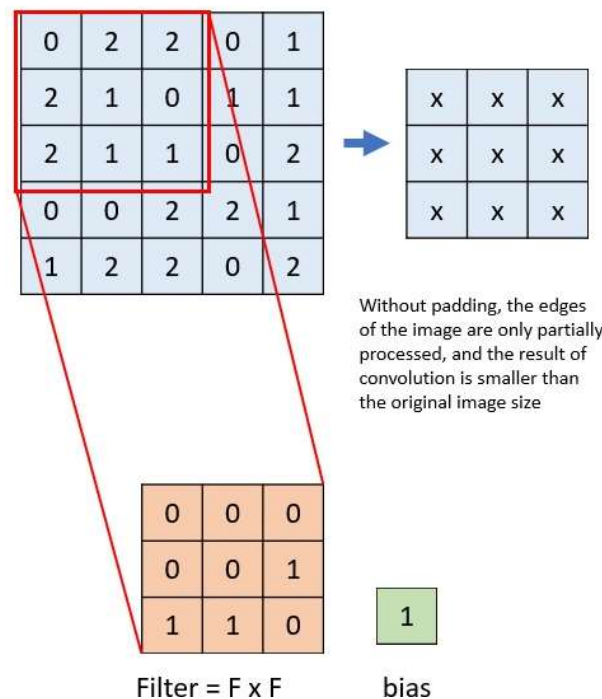
Intuition behind CNN:

Different Layers:

- Convolution layer

Lets take a look at sizes more in depth

$$Output\ size = \frac{W - k + 2P}{S} + 1$$



Intuition behind CNN:

Different Layers:

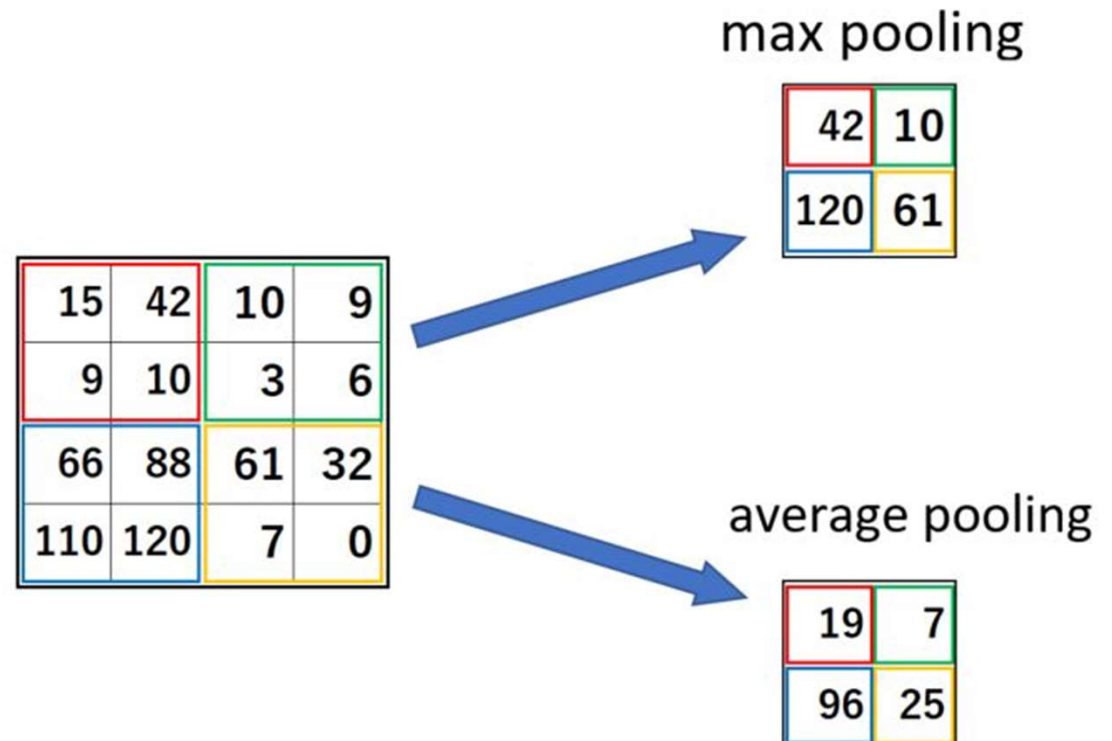
- Pooling layer

This layer will reduce the sample size (downsample)

- Max pooling
- Average pooling

Size of the output of a pooling layer

$$\text{Output size} = \frac{W - \text{pool size}}{S} + 1$$



Intuition behind CNN:

Different Layers:

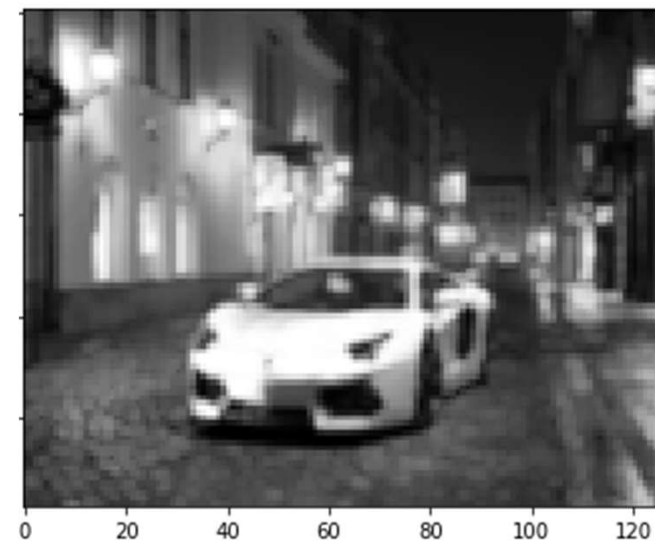
- example



Original Image



Convolved image



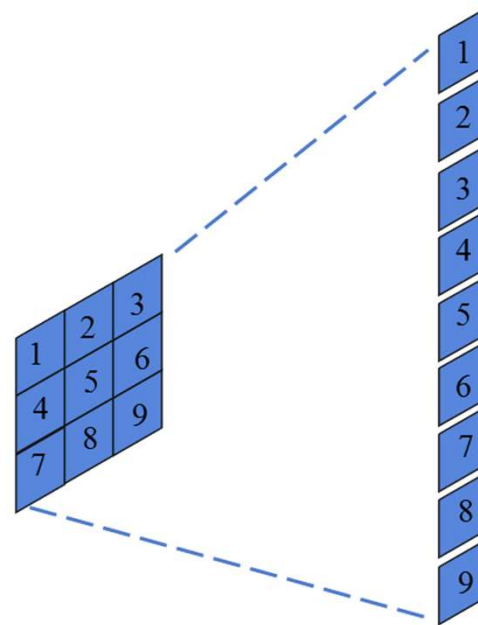
MAX Pooling

Intuition behind CNN:

Different Layers:

- Flatten layer

This will flatten the 2D array into a 1D

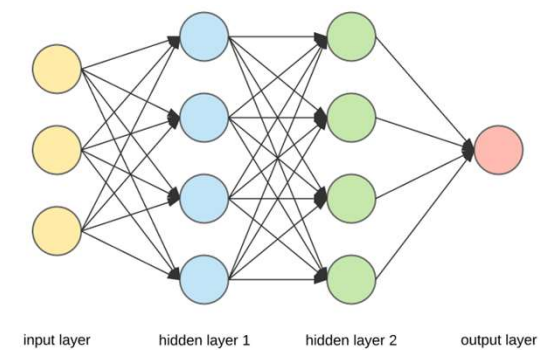
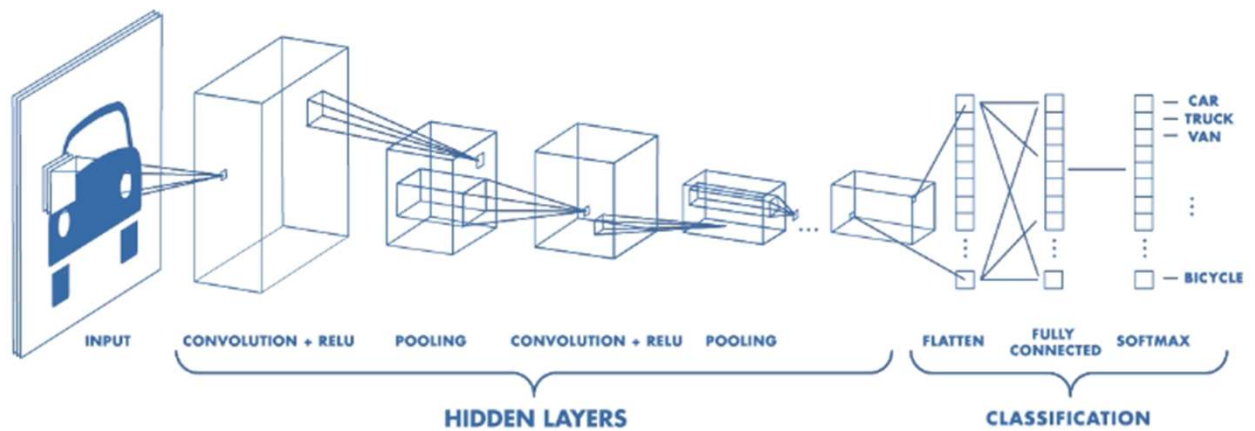


Intuition behind CNN:

Different Layers:

- Fully connected layer

This is simply a ANN attached to the end of previous layers:



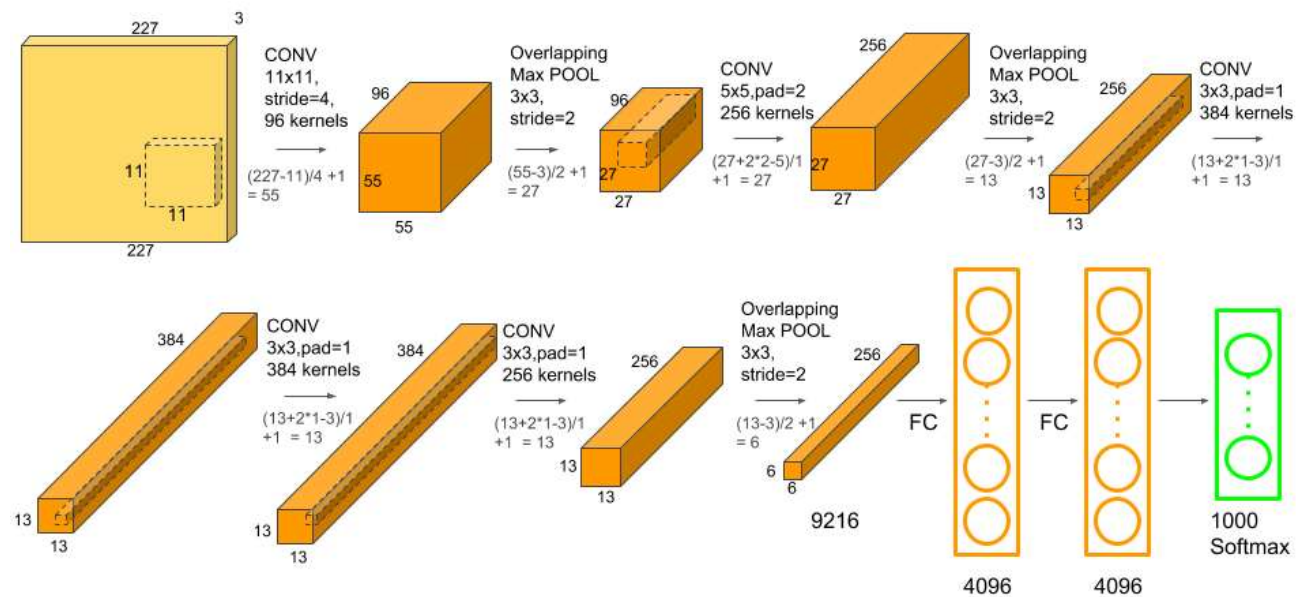
Intuition behind CNN:

Different Layers:

- Let's see how sizes change as we proceed through a CNN using an example:

For details info:

<https://www.learnopencv.com/number-of-parameters-and-tensor-sizes-in-convolutional-neural-network/>



Intuition behind CNN:

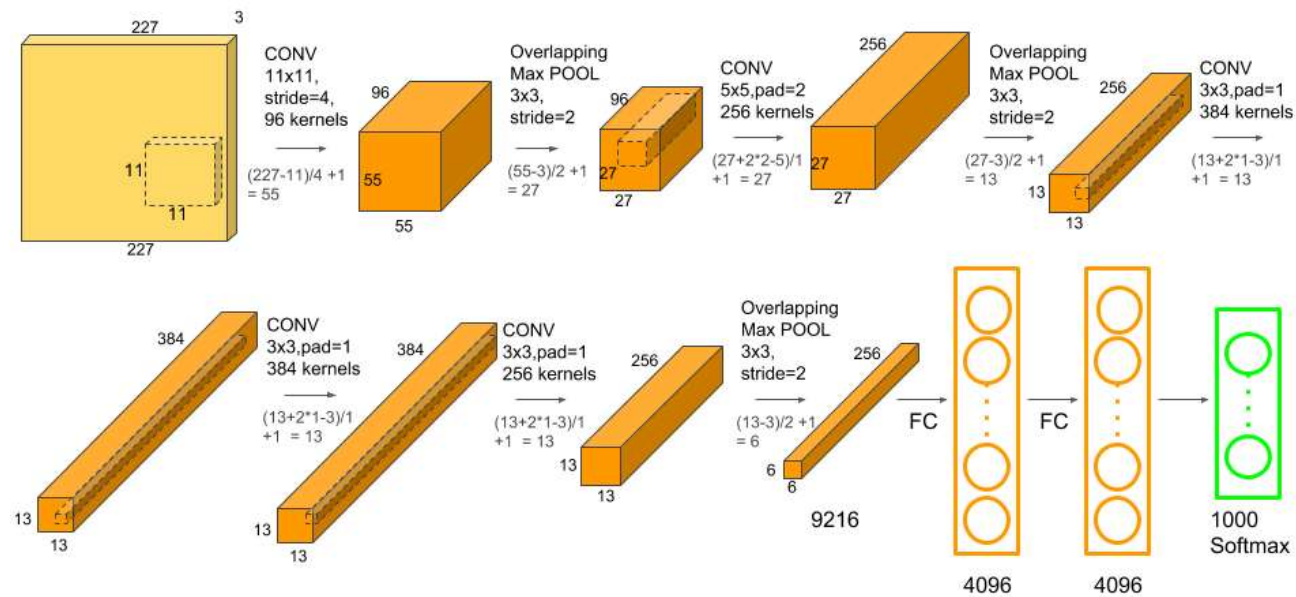
Different Layers:

- Lets see how sizes changes as we proceed through a CNN using an example:
- Input: $227 \times 227 \times 3$
- Apply 96 (11×11) kernels: $S=1$, $P=0$, Next Conv Layer will be $55 \times 55 \times 96$

$$\text{Output size} = \frac{W - k + 2P}{S} + 1 = 55$$

- Max pool (3×3), $S=2$

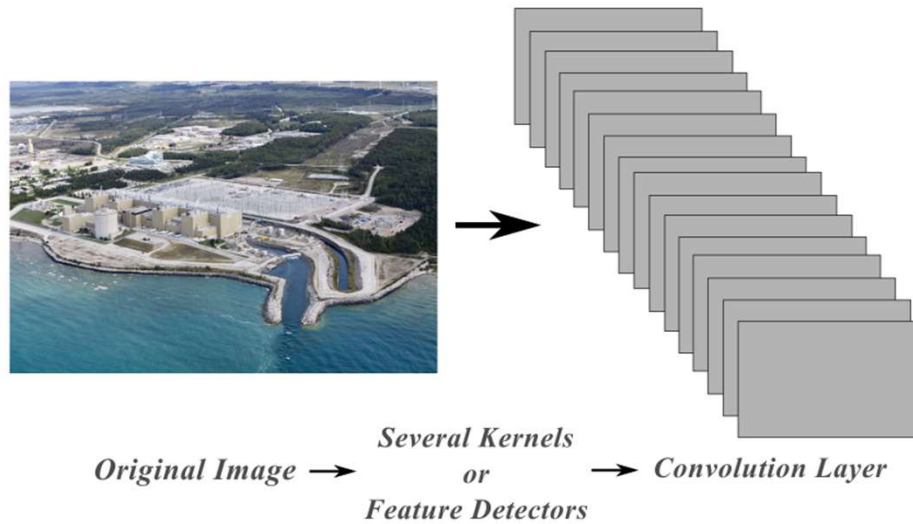
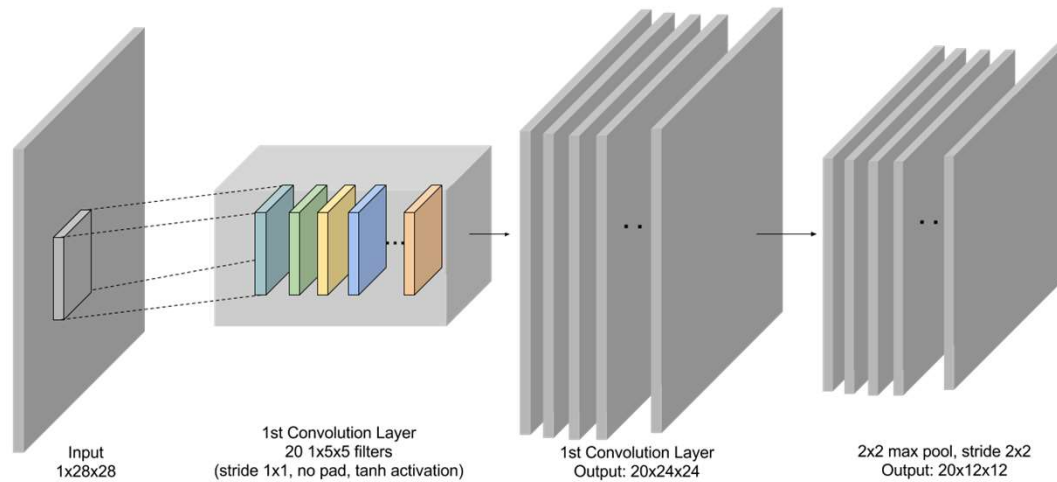
$$\text{Output size} = \frac{W - \text{pool size}}{S} + 1 = 27$$



Intuition behind CNN:

Different Layers:

- Another example





Python:

Hands on exercise:

Fashion MINST dataset

Dataset: <https://github.com/zalandoresearch/fashion-mnist>

OR: <https://www.kaggle.com/zalando-research/fashionmnist>



Python:

Fashion MINST dataset has 10 classes:

Label	Class
0	T-shirt/top
1	Trouser
2	Pullover
3	Dress
4	Coat
5	Sandal
6	Shirt
7	Sneaker
8	Bag
9	Ankle boot





Python:

Hands on exercise:

As an another exercise you can try to build a CNN that classifies MNIST handwritten digits



Python:

Hands on exercise:

Build a CNN that classifies pictures of humans into 2 categories:

People in the photo have helmet

People in the photo don't have helmet

